

Muhammad Abu Bakkar Aslam

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PROFILE

BSIT Graduate and aspiring Machine Learning Engineer with hands-on experience in Python, Scikit-learn, and data preprocessing. Skilled in building, optimizing, and evaluating machine learning models using regression, classification, ensemble methods, hyperparameter tuning, cross-validation, and ML pipelines. Passionate about solving real-world problems through data-driven solutions.

EDUCATION

Bachelor of Science in Information Technology

Lahore Garrison University

2022 – 2026

Lahore, Pakistan

Intermediate (FSc)

Garrison College for Boys Lahore Cantt

2019 – 2021

Lahore, Pakistan

TECHNICAL SKILL

- **Programming Languages:** Python
- **Machine Learning:** Scikit-learn, Pandas, NumPy, Matplotlib, Regression, Classification, Decision Trees, Random Forest, Naive Bayes, KNN, SVM, XGBoost, CatBoost, Model Evaluation, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Pipelines
- **Data Processing:** Data Cleaning, Data Preprocessing, Feature Scaling, Label Encoding, One-Hot Encoding, Missing Value Handling, Exploratory Data Analysis (EDA)
- **Tools & Technologies:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Anaconda mobile app screens applications Collaborated on user-centered design concepts; applied **HCI principles** to improve usability, accessibility, and operator efficiency

PROJECTS

House Price Prediction | Multiple Regression + ML Optimization - *Python, Pandas, NumPy, Scikit-Learn, Seaborn*

- Built a regression model to predict house prices using property features.
- Applied preprocessing techniques including encoding and feature scaling.
- Compared multiple regression models and evaluated performance using **R² Score** and **MAE**.
- Optimized model performance using cross-validation and hyperparameter tuning.

Titanic Survival Prediction | Classification Project- *Python, Pandas, NumPy, Scikit-Learn, Seaborn*

- Developed classification models to predict passenger survival.
- Performed missing value handling, feature encoding, and feature selection.
- Implemented Logistic Regression, Decision Tree, Random Forest, and Naive Bayes.
- Evaluated models using Accuracy, Precision, Recall, F1 Score, and Confusion Matrix

Smart Protective Socket (Final Year Project) - *ESP32, Sensors, Relay, Blynk IoT*

- Designed an IoT-based electrical safety device to detect human contact with live current.
- Integrated ESP32 microcontroller, sensors, relay module, and Blynk IoT platform.
- Developed automatic power cutoff mechanism and emergency notification system.
- Focused on preventing electrical accidents and improving household safety.

COMMUNICATION

- Urdu
- English